📍 SLAM-Toolbox Pure Localization and Autonomous Navigation

1. Course Content

- Master using SLAM-Toolbox to load offline pose graph (`.posegraph` + `.data`) for high-precision global pure localization (Pure Localization Mode).
- Combine SLAM-Toolbox pure localization pose output with Nav2 navigation framework (TEB local trajectory optimization planner) to achieve autonomous obstacle-avoiding single-point/multi-point navigation in known map environments.
- Understand the collaborative workflow of graph optimization sliding window global localization and Nav2 path planning stack in non-AMCL mode.

---

2. SLAM-Toolbox Pure Localization Navigation Principle Analysis

In actual industrial and research applications, after map building is complete, there is usually no need to continue incrementally expanding the map during navigation. SLAM-Toolbox provides a dedicated pure localization mode (Localization Mode):

1. **Graph Structure Freezing and Sliding Window Optimization (Sliding Window)**:
   After loading offline pose graph (`.posegraph` and `.data`), the system uses it as read-only prior, no longer inserting permanent new nodes into the global pose graph; only performs real-time nonlinear optimization of the robot's current pose within a very small sliding window (`scan_buffer_size: 3`) using the Ceres solver, greatly reducing CPU and memory load.
2. **Advantages Over Traditional AMCL Particle Filter**:
   Traditional AMCL relies on Monte Carlo particle filter, which is prone to particle dispersion degradation, jumping, or losing track (Pose Jump) in long corridors or symmetric feature environments; SLAM-Toolbox, based on laser point cloud and dense pose graph matching, outputs high-frequency smooth, jitter-free `map -> odom` coordinate transformation.
3. **Seamless Nav2 Navigation Protocol Stack Integration**:
   The system shields native AMCL node through modified `slamtoolx_bringup.launch.py`, with `localization_slam_toolbox_node` solely responsible for publishing global localization, while Nav2 focuses on costmap 2D updates, global path planning (NavfnPlanner), and TEB local trajectory obstacle avoidance control.

---

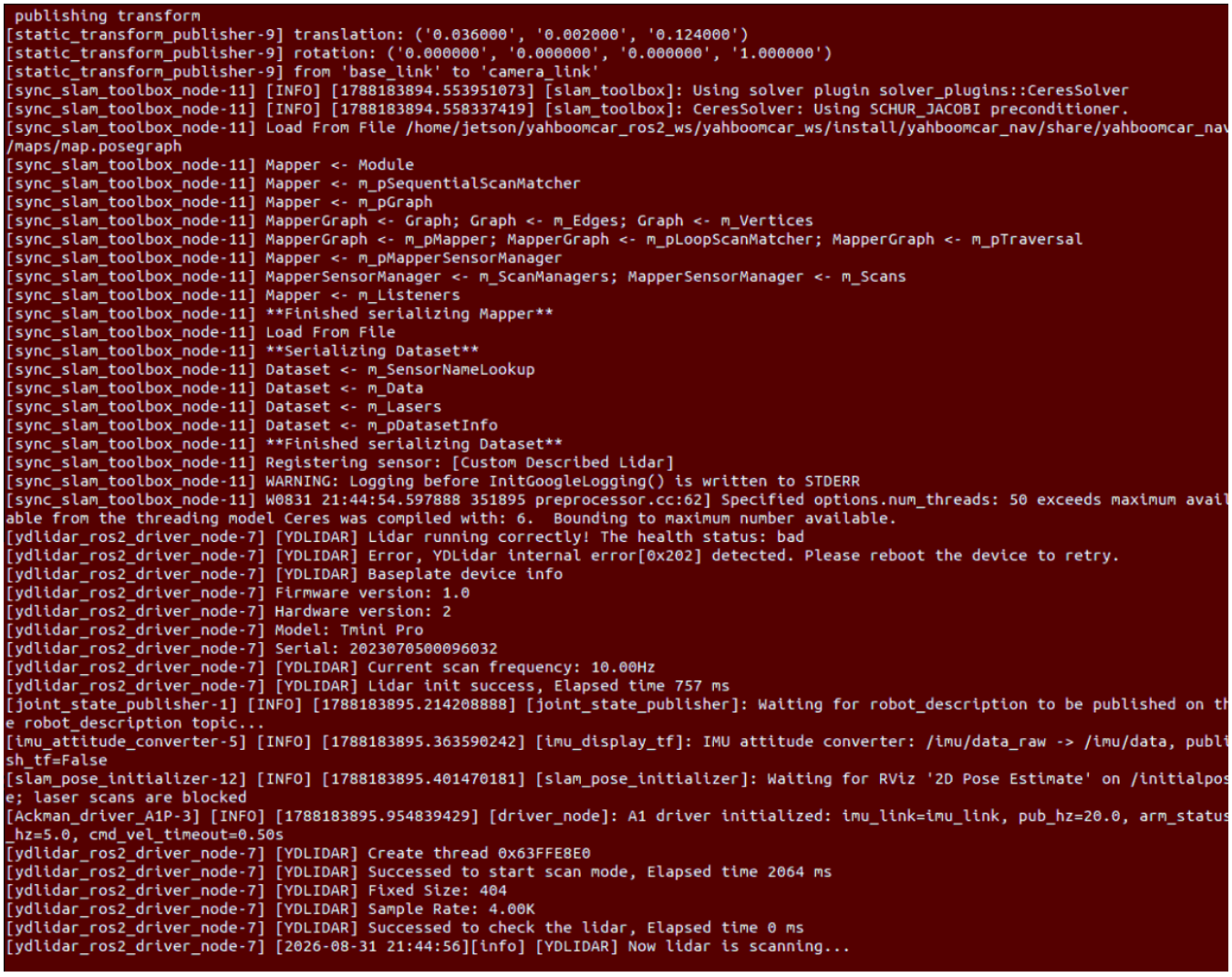
3. Preparation and Running

This example requires opening 3 terminals to start the following nodes in sequence:

3.1 Start Chassis + Lidar

Run in **Terminal 1** (start A1 Pro chassis driver, wheel odometry, and lidar driver):

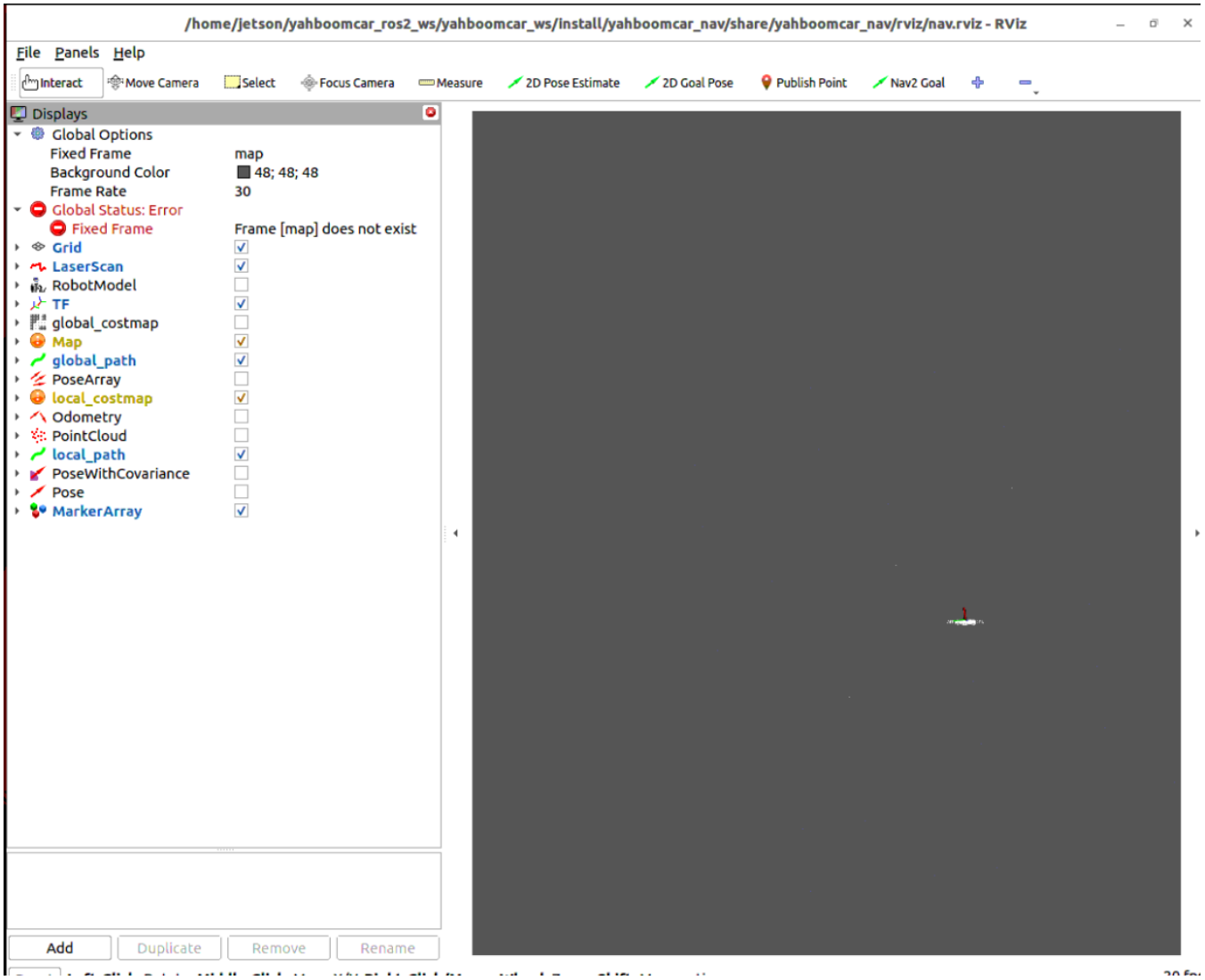
```
ros2 launch yahboomcar_nav laser_bringup_launch.py
```



3.2 Start RViz Navigation Visualization

Run in **Terminal 2** (it is recommended to open RViz on the virtual machine if the VM and robot are on the same LAN):

```
ros2 launch yahboomcar_nav display_nav_launch.py
```

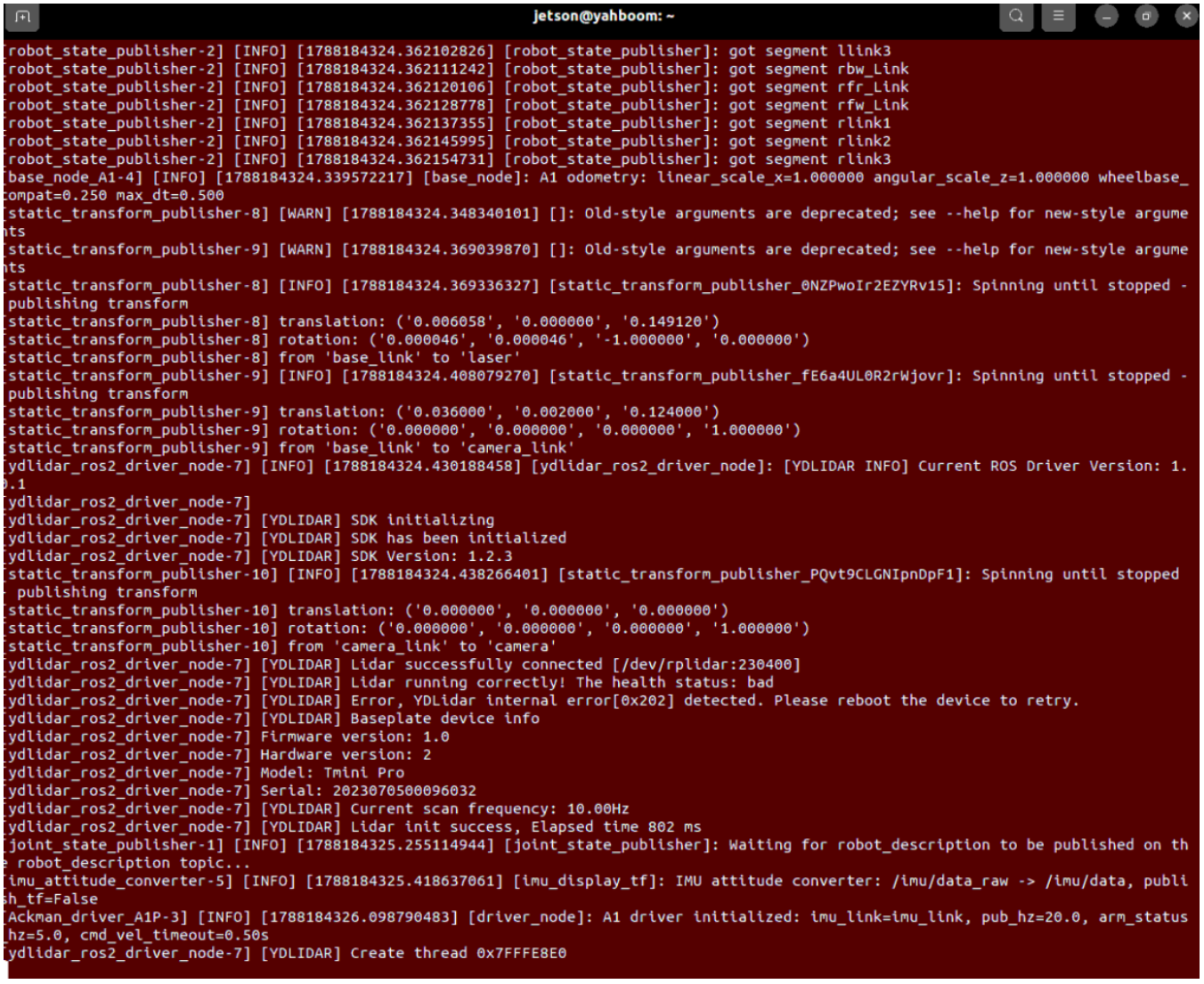


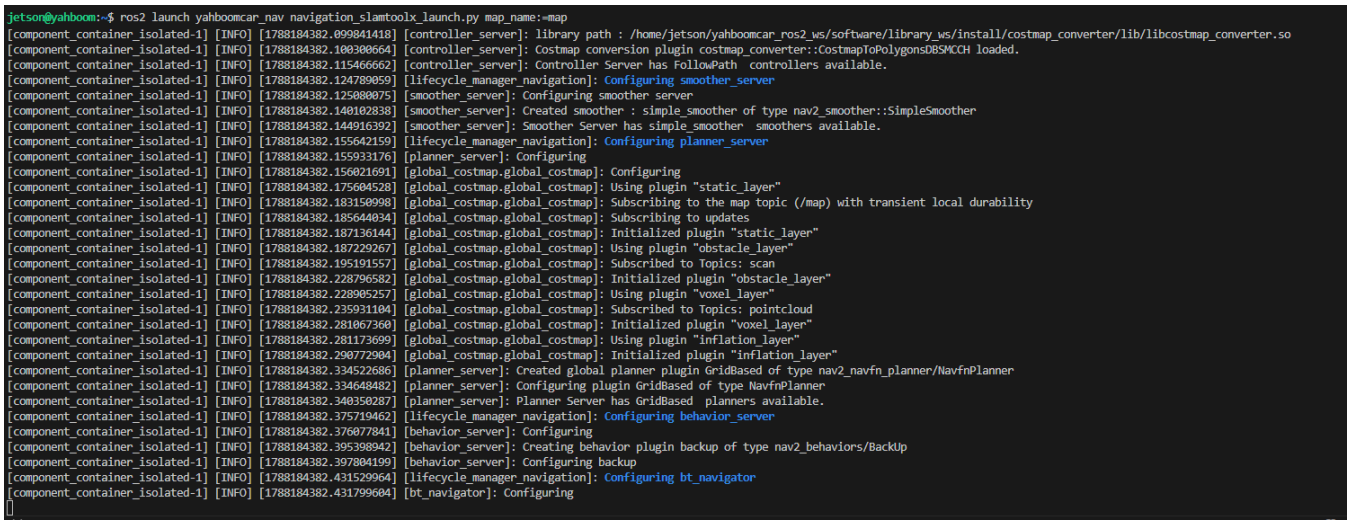
3.3 Start SLAM-Toolbox Pure Localization Navigation Core

Run in **Terminal 3** (one-click pull up SLAM-Toolbox pure localization node and Nav2 path planning stack):

```
ros2 launch yahboomcar_nav navigation_slamtoolx_launch.py map:=map.yaml pose_map:=map
```

> **Tip**
>
> - `map`: Specifies the 2D raster map configuration file to load (stored in `yahboomcar_nav/maps/` directory).
> - `pose_map`: Specifies the SLAM-Toolbox pose graph prefix to load (corresponds to `yahboomcar.posegraph` and `yahboomcar.data`).
> - If using a custom named map (for example `map`), you can pass: `map:=map.yaml pose_map:=map`.

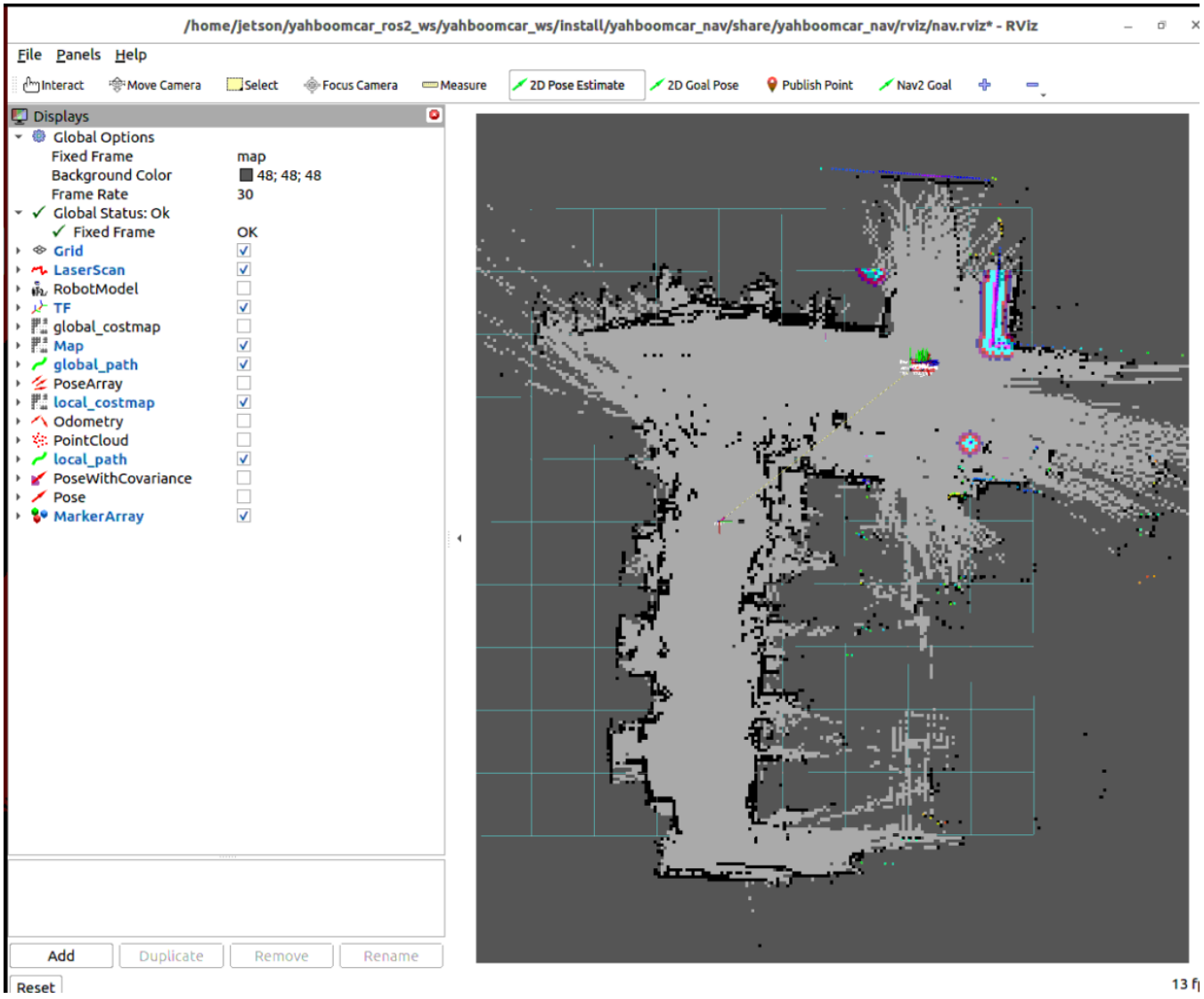

jetson@yahboom: ~
robot_state_publisher-2] [INFO] [1788184324.362102826] [robot_state_publisher]: got segment llink3
robot_state_publisher-2] [INFO] [1788184324.362111242] [robot_state_publisher]: got segment rbw_Link
robot_state_publisher-2] [INFO] [1788184324.362120106] [robot_state_publisher]: got segment rfr_Link
robot_state_publisher-2] [INFO] [1788184324.362128778] [robot_state_publisher]: got segment rfw_Link
robot_state_publisher-2] [INFO] [1788184324.362137355] [robot_state_publisher]: got segment rlink1
robot_state_publisher-2] [INFO] [1788184324.362145995] [robot_state_publisher]: got segment rlink2
robot_state_publisher-2] [INFO] [1788184324.362154731] [robot_state_publisher]: got segment rlink3
base_node_A1-4] [INFO] [1788184324.339572217] [base_node]: A1 odometry: linear_scale_x=1.000000 angular_scale_z=1.000000 wheelbase_
compat=0.250 max_dt=0.500
static_transform_publisher-8] [WARN] [1788184324.348340101] []: Old-style arguments are deprecated; see --help for new-style argume
nts
static_transform_publisher-9] [WARN] [1788184324.369039870] []: Old-style arguments are deprecated; see --help for new-style argume
nts
static_transform_publisher-8] [INFO] [1788184324.369336327] [static_transform_publisher_0NZPwoIr2EZYRv15]: Spinning until stopped -
publishing transform
static_transform_publisher-8] translation: ('0.006058', '0.000000', '0.149120')
static_transform_publisher-8] rotation: ('0.000046', '0.000046', '-1.000000', '0.000000')
static_transform_publisher-8] from 'base_link' to 'laser'
static_transform_publisher-9] [INFO] [1788184324.408079270] [static_transform_publisher_fE6a4UL0R2rWjovr]: Spinning until stopped -
publishing transform
static_transform_publisher-9] translation: ('0.036000', '0.002000', '0.124000')
static_transform_publisher-9] rotation: ('0.000000', '0.000000', '0.000000', '1.000000')
static_transform_publisher-9] from 'base_link' to 'camera_link'
ydlidar_ros2_driver_node-7] [INFO] [1788184324.430188458] [ydlidar_ros2_driver_node]: [YDLIDAR INFO] Current ROS Driver Version: 1.
0.1
ydlidar_ros2_driver_node-7]
ydlidar_ros2_driver_node-7] [YDLIDAR] SDK initializing
ydlidar_ros2_driver_node-7] [YDLIDAR] SDK has been initialized
ydlidar_ros2_driver_node-7] [YDLIDAR] SDK Version: 1.2.3
static_transform_publisher-10] [INFO] [1788184324.438266401] [static_transform_publisher_PQvt9CLGNIpnDpF1]: Spinning until stopped
- publishing transform
static_transform_publisher-10] translation: ('0.000000', '0.000000', '0.000000')
static_transform_publisher-10] rotation: ('0.000000', '0.000000', '0.000000', '1.000000')
static_transform_publisher-10] from 'camera_link' to 'camera'
ydlidar_ros2_driver_node-7] [YDLIDAR] Lidar successfully connected [/dev/rplidar:230400]
ydlidar_ros2_driver_node-7] [YDLIDAR] Lidar running correctly! The health status: bad
ydlidar_ros2_driver_node-7] [YDLIDAR] Error, YDLidar internal error[0x202] detected. Please reboot the device to retry.
ydlidar_ros2_driver_node-7] [YDLIDAR] Baseplate device info
ydlidar_ros2_driver_node-7] Firmware version: 1.0
ydlidar_ros2_driver_node-7] Hardware version: 2
ydlidar_ros2_driver_node-7] Model: Tmini Pro
ydlidar_ros2_driver_node-7] Serial: 2023070500096032
ydlidar_ros2_driver_node-7] [YDLIDAR] Current scan frequency: 10.00Hz
ydlidar_ros2_driver_node-7] [YDLIDAR] Lidar init success, Elapsed time 802 ms
joint_state_publisher-1] [INFO] [1788184325.255114944] [joint_state_publisher]: Waiting for robot_description to be published on th
e robot_description topic...
imu_attitude_converter-5] [INFO] [1788184325.418637061] [imu_display_tf]: IMU attitude converter: /imu/data_raw -> /imu/data, publi
sh_tf=False
Ackman_driver_A1P-3] [INFO] [1788184326.098790483] [driver_node]: A1 driver initialized: imu_link=imu_link, pub_hz=20.0, arm_status
_hz=5.0, cmd_vel_timeout=0.50s
ydlidar_ros2_driver_node-7] [YDLIDAR] Create thread 0x7FFFE8E0


jetson@yahboom:~$ ros2 launch yahboomcar_nav navigation_slamtoolx_launch.py map_name:=map
[component_container_isolated-1] [INFO] [1788184382.099841418] [controller_server]: library path : /home/jetson/yahboomcar_ros2_ws/software/library_ws/install/costmap_converter/lib/libcostmap_converter.so
[component_container_isolated-1] [INFO] [1788184382.100300664] [controller_server]: Costmap conversion plugin costmap_converter::CostmapToPolygonsDBSMCCH loaded.
[component_container_isolated-1] [INFO] [1788184382.115466662] [controller_server]: Controller Server has FollowPath controllers available.
[component_container_isolated-1] [INFO] [1788184382.124789059] [lifecycle_manager_navigation]: Configuring smoother_server
[component_container_isolated-1] [INFO] [1788184382.125080075] [smoother_server]: Configuring smoother server
[component_container_isolated-1] [INFO] [1788184382.140102838] [smoother_server]: Created smoother : simple_smoother of type nav2_smoother::SimpleSmoother
[component_container_isolated-1] [INFO] [1788184382.144916392] [smoother_server]: Smoother Server has simple_smoother smoothers available.
[component_container_isolated-1] [INFO] [1788184382.155642159] [lifecycle_manager_navigation]: Configuring planner_server
[component_container_isolated-1] [INFO] [1788184382.155933176] [planner_server]: Configuring
[component_container_isolated-1] [INFO] [1788184382.156021691] [global_costmap.global_costmap]: Configuring
[component_container_isolated-1] [INFO] [1788184382.175604528] [global_costmap.global_costmap]: Using plugin "static_layer"
[component_container_isolated-1] [INFO] [1788184382.183150998] [global_costmap.global_costmap]: Subscribing to the map topic (/map) with transient local durability
[component_container_isolated-1] [INFO] [1788184382.185644034] [global_costmap.global_costmap]: Subscribing to updates
[component_container_isolated-1] [INFO] [1788184382.187136144] [global_costmap.global_costmap]: Initialized plugin "static_layer"
[component_container_isolated-1] [INFO] [1788184382.187229267] [global_costmap.global_costmap]: Using plugin "obstacle_layer"
[component_container_isolated-1] [INFO] [1788184382.195191557] [global_costmap.global_costmap]: Subscribed to Topics: scan
[component_container_isolated-1] [INFO] [1788184382.228796582] [global_costmap.global_costmap]: Initialized plugin "obstacle_layer"
[component_container_isolated-1] [INFO] [1788184382.228905257] [global_costmap.global_costmap]: Using plugin "voxel_layer"
[component_container_isolated-1] [INFO] [1788184382.235931104] [global_costmap.global_costmap]: Subscribed to Topics: pointcloud
[component_container_isolated-1] [INFO] [1788184382.281067360] [global_costmap.global_costmap]: Initialized plugin "voxel_layer"
[component_container_isolated-1] [INFO] [1788184382.281173699] [global_costmap.global_costmap]: Using plugin "inflation_layer"
[component_container_isolated-1] [INFO] [1788184382.290772904] [global_costmap.global_costmap]: Initialized plugin "inflation_layer"
[component_container_isolated-1] [INFO] [1788184382.334522686] [planner_server]: Created global planner plugin GridBased of type nav2_navfn_planner/NavfnPlanner
[component_container_isolated-1] [INFO] [1788184382.334648482] [planner_server]: Configuring plugin GridBased of type NavfnPlanner
[component_container_isolated-1] [INFO] [1788184382.340350287] [planner_server]: Planner Server has GridBased planners available.
[component_container_isolated-1] [INFO] [1788184382.375719462] [lifecycle_manager_navigation]: Configuring behavior_server
[component_container_isolated-1] [INFO] [1788184382.376077841] [behavior_server]: Configuring
[component_container_isolated-1] [INFO] [1788184382.395398942] [behavior_server]: Creating behavior plugin backup of type nav2_behaviors/BackUp
[component_container_isolated-1] [INFO] [1788184382.397804199] [behavior_server]: Configuring backup
[component_container_isolated-1] [INFO] [1788184382.431529964] [lifecycle_manager_navigation]: Configuring bt_navigator
[component_container_isolated-1] [INFO] [1788184382.431799604] [bt_navigator]: Configuring

4. Relocalization Navigation Operation Flow and Effect Description

4.1 Initial Relocalization Calibration (2D Pose Estimate)

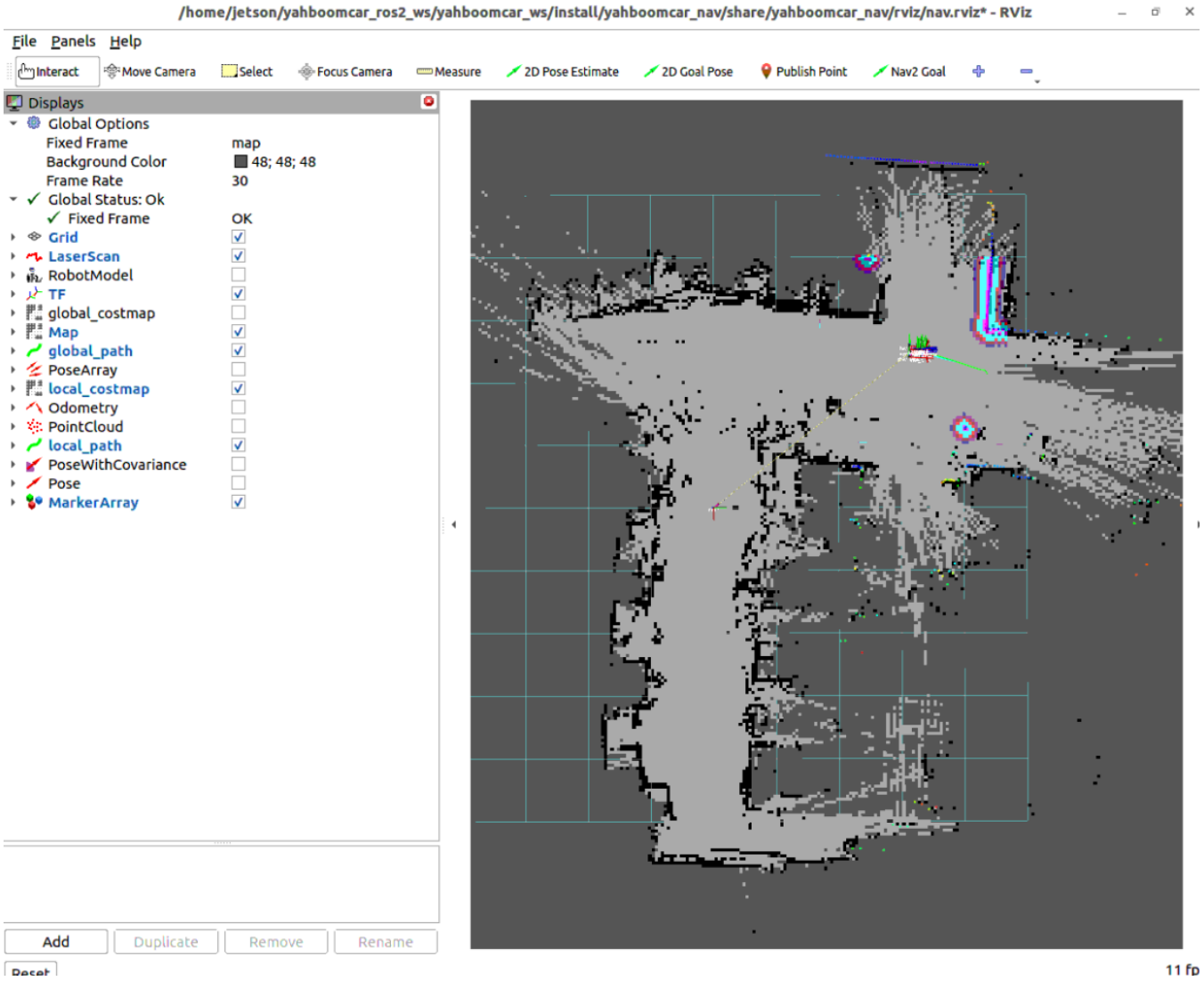
1. Click the `2D Pose Estimate` tool in the top toolbar of RViz2;
2. Click at the approximate actual robot stopping position on the map, and hold and drag the mouse along the chassis heading direction to release the green arrow to publish initial pose estimate;
3. After SLAM-Toolbox receives the pose, it will immediately complete local matching and alignment with surrounding geometric features; observe that the red laser point cloud in RViz completely coincides with the black obstacle contours on the map, representing successful relocalization.



4.2 Single-Point Autonomous Obstacle-Avoiding Navigation (Nav2 Goal)

1. Click the `Nav2 Goal` (or `2D Goal Pose`) tool in the top toolbar of RViz2;
2. Click in the free traversable area on the map and drag out the expected goal heading arrow;
3. Nav2 global planner will generate an optimal path, local TEB planner real-time controls Ackermann

chassis front-wheel steering and vehicle speed, dynamically avoids obstacles on the travel route, and smoothly arrives at the goal point.



4.3 Multi-Point Autonomous Continuous Cruising (Waypoint Following)

If you need the robot to continuously cruise multiple goal points:

1. In the **Navigation2** panel in the lower-left corner of RViz2 interface, switch the mode from `Navigate to Pose` to `Waypoint / Nav Through Poses Mode`;
2. Use the `Nav2 Goal` tool to sequentially click and calibrate multiple target waypoints on the map;
3. Click `Start Waypoint Following`, and the robot will autonomously travel through all waypoints in the set sequence and steadily align and stop at each point.

> **Note**
>
> To end navigation task, press `Ctrl+C` in each terminal window to stop the program.

5. Core Source Code and Localization Parameter Analysis

5.1 Pure Localization and Navigation Integration Launch Analysis

1. Main Launch Scheduling Launch

File Path:

```
$HOME/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/launch/nav2/navigation_slamtoolx_launch.py
```

```python
def generate_launch_description():
    ...
    # Map and parameter configuration
    map_dir = PathJoinSubstitution([map_folder_dir, LaunchConfiguration("map")])
    pose_map = PathJoinSubstitution([map_folder_dir, LaunchConfiguration("pose_map")])
    params_file = LaunchConfiguration("params_file")

    # 1. Start Navigation2 stack (using modified slamtoolx_bringup to shield AMCL)
    nav2_bringup = IncludeLaunchDescription(
        PythonLaunchDescriptionSource(nav2_launch_mod),
        launch_arguments={
            "map": map_dir,
            "params_file": params_file,
            "use_sim_time": use_sim_time,
        }.items()
    )

    # 2. Start SLAM-Toolbox pure localization node
    slamtoolbox_localization = IncludeLaunchDescription(
        PythonLaunchDescriptionSource(slamtoolbox_localization_launch_file),
        launch_arguments={
            "map": pose_map,
            "use_sim_time": use_sim_time,
        }.items()
    )

    return LaunchDescription([
        *declared_arguments,
        slamtoolbox_localization_nav,
    ])
```

2. SLAM-Toolbox Pure Localization Node Launch

File Path:

```
$HOME/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/launch/nav2/localization_slamtoolx_launch.py
```

```python
# Dynamically replace pose graph path through RewrittenYaml and start localization_slam_toolbox_node
```

```
configured_params = RewrittenYaml(
    source_file=slam_params_file,
    root_key='',
    param_rewrites={
        'use_sim_time': use_sim_time,
        'map_file_name': map_file,
    },
    convert_types=True
)

slam_toolbox_node = Node(
    package='slam_toolbox',
    executable='localization_slam_toolbox_node',
    name='slam_toolbox',
    output='screen',
    parameters=[configured_params],
)
```

5.2 Core Algorithm and Navigation Parameter Interpretation

Parameter Configuration File Paths:

```
$HOME/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/params/mapper_params_localization.yaml
$HOME/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/params/teb_nav_params.yaml
```

Core Parameter Description Table

Parameter Module	Parameter Name	Current Config Value	Core Function and Physical Meaning
SLAM-Toolbox Localization	`mode`	`localization`	Working mode set to pure localization (does not add permanent new nodes)
SLAM-Toolbox Localization	`scan_buffer_size`	`3`	Shortened sliding window scan buffer to 3 frames, extremely low computational latency
SLAM-Toolbox Localization	`map_start_at_dock`	`true`	Whether to allow loading and alignment from map reference position
SLAM-Toolbox Localization	`distance_variance_penalty`	`0.5`	Scan matching displacement degradation penalty factor
SLAM-Toolbox Localization	`angle_variance_penalty`	`1.0`	Scan matching angle degradation penalty factor

Parameter Module	Parameter Name	Current Config Value	Core Function and Physical Meaning
Nav2 Local Controller	`FollowPath.plugin`	`TebLocalPlannerROS`	TEB local time-optimal trajectory optimization planner plugin
Nav2 Local Controller	`max_vel_x`	`0.25`	Maximum cruising linear velocity (m/s)
Nav2 Local Controller	`min_vel_x`	`-0.05`	Limit reverse linear velocity (prevent rear collision due to no lidar)
Nav2 Local Controller	`max_vel_theta`	`1.0`	Maximum turning angular velocity (rad/s)
Nav2 Goal Tolerance	`xy_goal_tolerance`	`0.25`	Goal point planar distance arrival tolerance (meters)
Nav2 Goal Tolerance	`yaw_goal_tolerance`	`0.25`	Goal point yaw angle arrival tolerance (radians ≈ 14.3°)
Nav2 Local Costmap	`inflation_radius`	`0.18`	Local obstacle dynamic inflation radius (meters)
Nav2 Local Costmap	`cost_scaling_factor`	`10.0`	Exponential scaling factor for obstacle cost decay with distance

Parameter File Core Configuration Snippet

```yaml
# 1. SLAM-Toolbox pure localization configuration (mapper_params_localization.yaml)
slam_toolbox:
  ros__parameters:
    solver_plugin: solver_plugins::CeresSolver
    odom_frame: odom
    map_frame: map
    base_frame: base_footprint
    scan_topic: /scan
    mode: localization              # Pure localization mode
    scan_buffer_size: 3             # Sliding window size is 3 frames
    resolution: 0.05
    min_laser_range: 0.05
    max_laser_range: 12.0
    do_loop_closing: true

# 2. TEB local trajectory optimization and obstacle avoidance configuration
(teb_nav_params.yaml)
controller_server:
  ros__parameters:
    controller_frequency: 20.0
    FollowPath:
      plugin: "teb_local_planner::TebLocalPlannerROS"
```

```yaml
      max_vel_x: 0.25                # Maximum cruising speed (m/s)
      min_vel_x: -0.05               # Limit reverse speed
      max_vel_theta: 1.0             # Maximum turning angular velocity (rad/s)
      min_obstacle_dist: 0.1         # Minimum obstacle hard distance (m)
      xy_goal_tolerance: 0.25        # Arrival distance accuracy (m)
      yaw_goal_tolerance: 0.25       # Arrival angle accuracy (rad)

local_costmap:
  local_costmap:
    ros__parameters:
      inflation_layer:
        plugin: "nav2_costmap_2d::InflationLayer"
        cost_scaling_factor: 10.0
        inflation_radius: 0.18       # Local inflation radius (m)
```